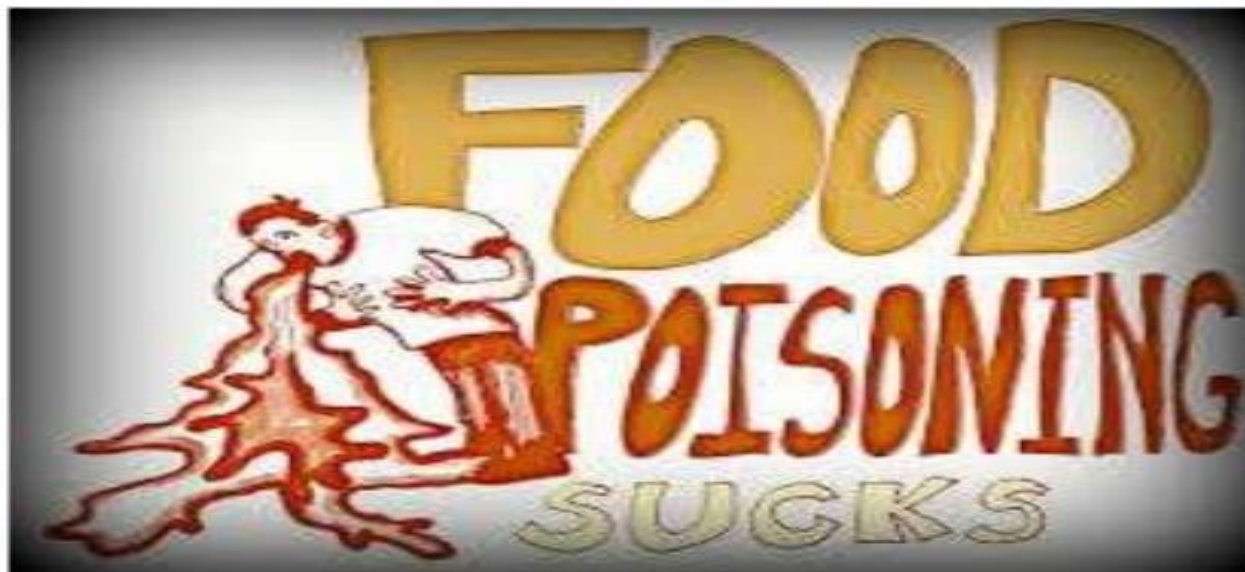




Assist.Prof.Dr.Vian B. Barzinji.
PhD.MSc. Medical Mycology

Objectives

- 1- Identify the most common causes for food poisoning.
- 2- Identify and describe the most common microorganisms involved in food poisoning.
- 3- Discuss how causing food poisoning.
- 4- Describe the ways to prevent food poisoning.



- According to the latest World Health Organization (WHO) data, unsafe food causes an estimated 866 million illnesses and 1.5 million deaths globally each year.

• Causes of Food Poisoning : Transmitted through food

- Unknown agents causes 68% of all food borne illness and related hospitalisations.



- Many cases of food poisoning are not reported.
- Main causes–
 - 1) Infectious agents include viruses, bacteria and parasites.
 - a. **Viruses** : Rotavirus, Hepatitis A virus
 - b. **Bacteria** : Salmonella, V. cholerae, E.coli, Botulism

Drinking contaminated water



Consuming stale or spoiled food



Eating street food



Eating undercooked meat or eggs



Eating unwashed fruits and vegetables



Cooking without washing hands



Poor food storage



Eating expired food



Not washing hands before eating food



Keeping raw and cooked food together



What Causes Food Poisoning?

c. **Parasites** : Trichinella spiralis

d. **Protozoan** : Entamoeba histolytica, Giardia lamblia

2) Other toxic agents, which includes

a). Chemicals: Pesticides , heavy metals, gasoline

b). **Fungi** : Mushrooms (*Amanita muscaria*)

c). Others : Puffer fish(*Fugu rubripes*), Ciguatera

3) Poor sanitation and preparation.

4) Improper packing and storage

What is food poisoning??

- Acute gastro enteritis acquired by ingestion of food and drink contaminated with pathogenic microorganisms (**bacteria, fungi ,viruses & protozoa,**) their **toxins & chemicals.**
- Food poisoning must be suspected when an acute illness with gastrointestinal or neurological manifestation affect two or more persons, who have shared a meal during the previous 72 hours.

Some common sources of Food poisoning



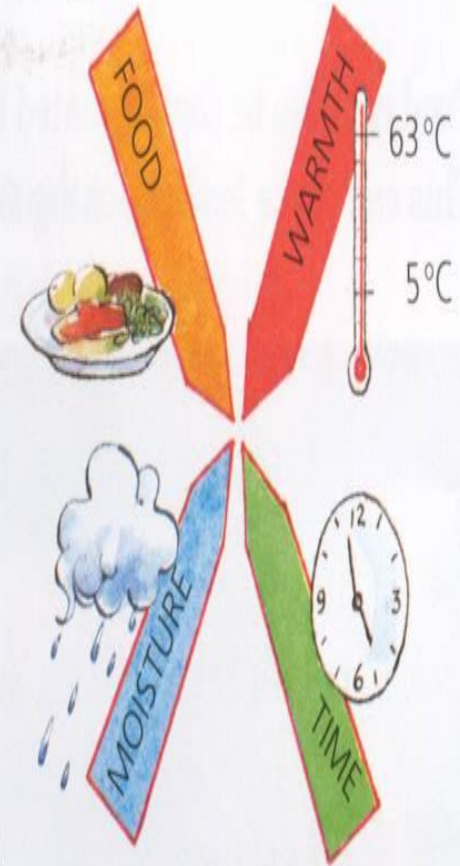
The food poisoning chain:-

There must be chain of events for food poisoning to occur.

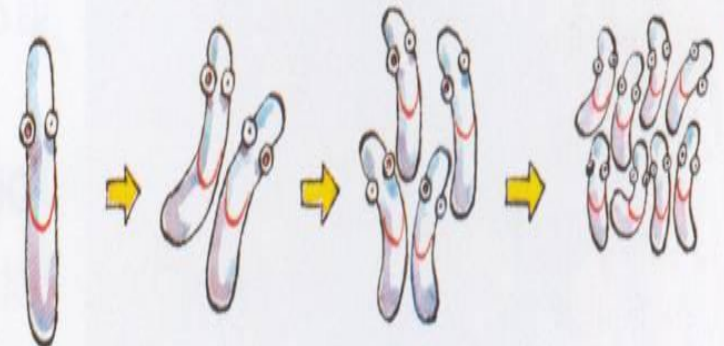
1. There must be bacteria on the food.
2. The bacteria must have the right conditions to grow that is warmth (between 5 C– 60C) and moisture.
3. The bacteria must have time to grow and multiply.

-By preventing each of these , the food poisoning chain will be broken and food poisoning prevented.

Bacteria need all four to multiply



Multiplication of bacteria



Characteristic of food poisoning

- 1- There is history of the ingestion of common food.
- Similarity of the sign and symptoms in the majority of the cases sharing meal.
- Attack of the many persons at the same time.

Even if you are young and healthy, you are at risk. pregnant women, infants, the elderly, and those weakened immune systems are at greater risk for foodborn illness

- Single cases of foodborne disease are difficult to identify, unless as in **botulism** there is distinctive clinical syndrome



Classification of food poisoning



1-Infection

Refers to the presence of bacteria or other microbes which infect the body after consumption ,
symptoms occur more slowly usually at least 24 hr)

2-Intoxication

Refers to ingestion of toxins contained, within the food, including bacteria produced exotoxin
symptom occur rapidly usually within 2-12 hr.)

What happens in your body?

- You consume contaminated food
- Bacteria passes through the stomach into the intestines
- Attaches to intestinal wall and multiplies
- Then either:
 - Stays attached
 - Travels to other body tissues
 - Produces a toxin that goes into your bloodstream
- The route the bacteria takes correlates to the symptoms a person would have.



E. coli

- Classification of food poisoning :

A. Based on symptoms and duration of onset.

B. Based on pathogenesis.

1. Food intoxications resulting from the ingestion of preformed bacterial toxins. (*Staphylococcus aureus*, *Bacillus cereus*, *Clostridium botulinum*, *Clostridium perfringens*)

2. Food intoxications caused by noninvasive bacteria that secrete toxins while adhering to the intestinal wall (*Enterotoxigenic E.coli*, *Vibrio cholerae*, *Campylobacter jejuni*)

3. Food intoxications that follow an intracellular invasion of the intestinal epithelial cells. (*Shigella*, *Salmonella*)

4. Diseases caused by bacteria that enter the blood stream via the intestinal tract. (*Salmonella typhi*, *Listeria monocytogenes*)

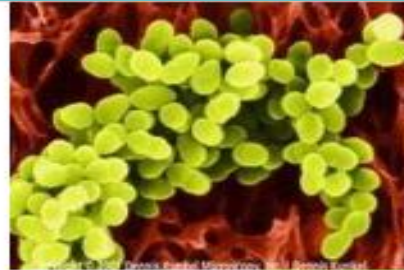
Food Poisoning Symptoms



Most common food poisoning bacteria



Salmonella
Found in animals,
raw poultry and
birds



**Staphylococcus
Aureus**
Found in human
nose and throat
(also skin)



**Clostridium
Perfringens**
Found in animals
and birds



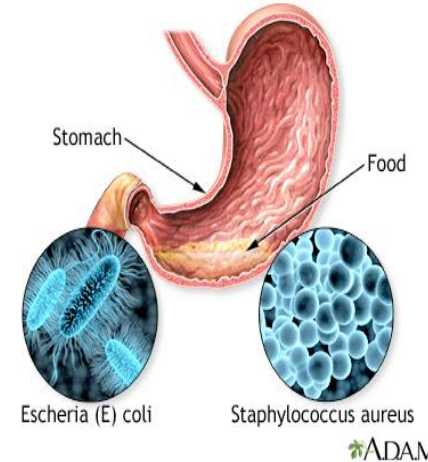
Bacillus Cereus
Found in soil,
vegetation,
cereals and
spices



**Clostridium
Botulinum**
Found in the soil
and associated
with vegetables
and meats

Staphylococcal food poisoning :

- Toxins of *Staphylococcus aureus*, which is a normal member of the microbiome of the skin and upper respiratory system, cause staphylococcal food poisoning. This is a common type of bacterial food poisoning.

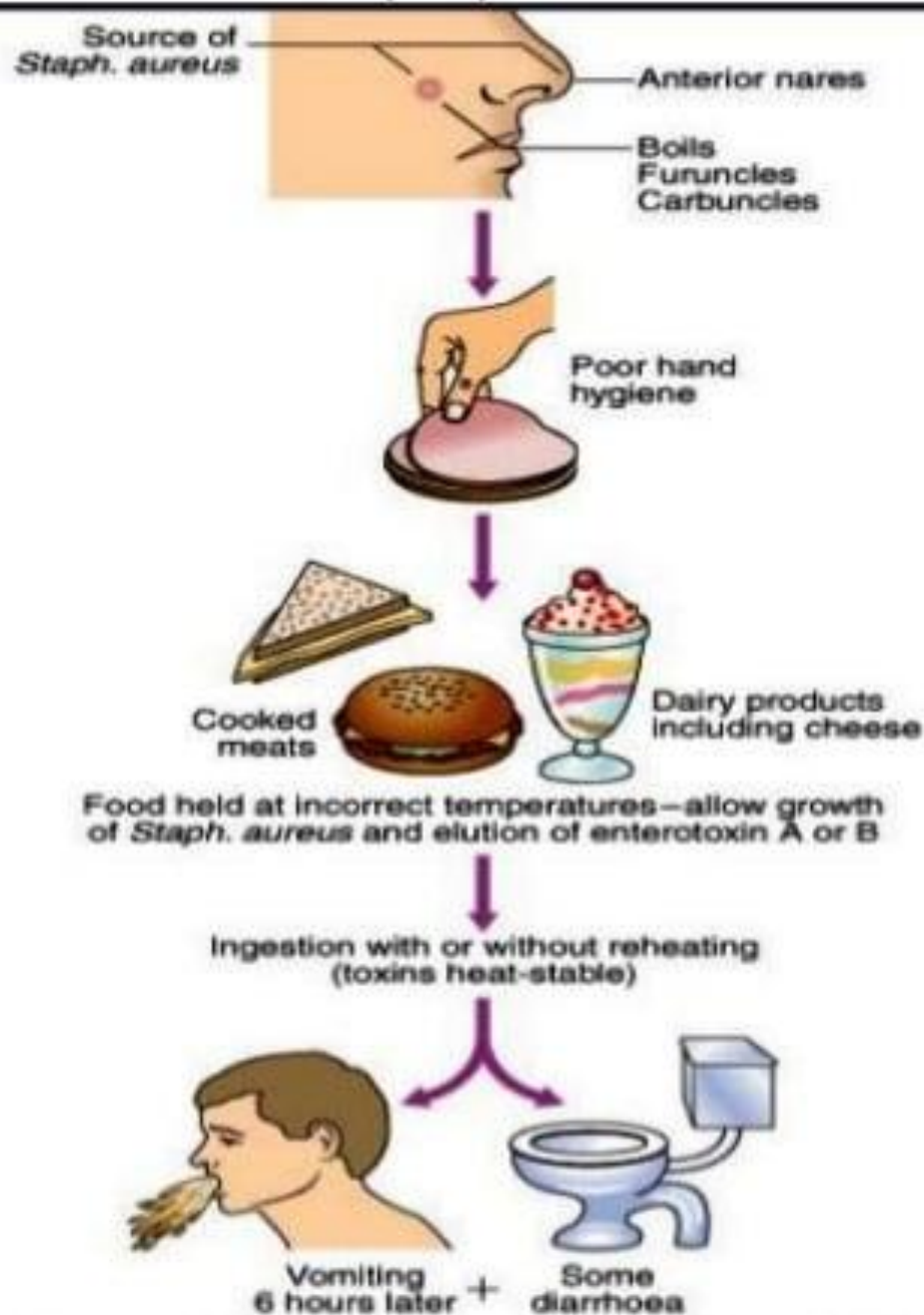


- **Only those strains that produce enterotoxin can cause food poisoning.**
- **Food is usually contaminated from infected food handler . The food handler with an active lesion can contaminate food.**
- **When foods remain at room temperature for several hours before being eaten, toxin producing Staphylococci multiply and elaborate the heat- stable toxin.**

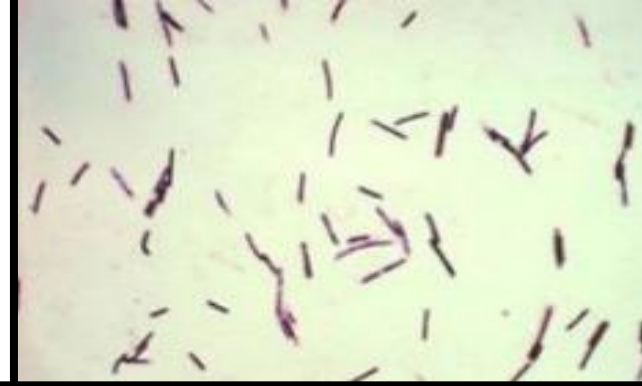
PATHOGENESIS:-

- Six heat stable (A-F).
- Binds to receptor in bowel, with stimulation medullary centers, vomiting follows.
- Staphylococcal enterotoxin are very stable their toxicity and antigenicity are not completely destroyed by boiling even canning.

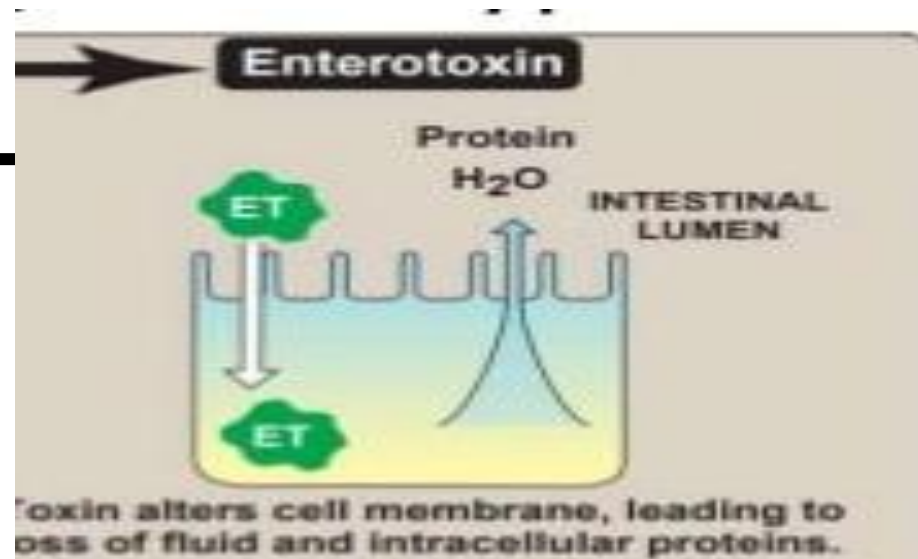
Pathogenesis of Staphylococcal enterotoxin.



Clostridium perfringens

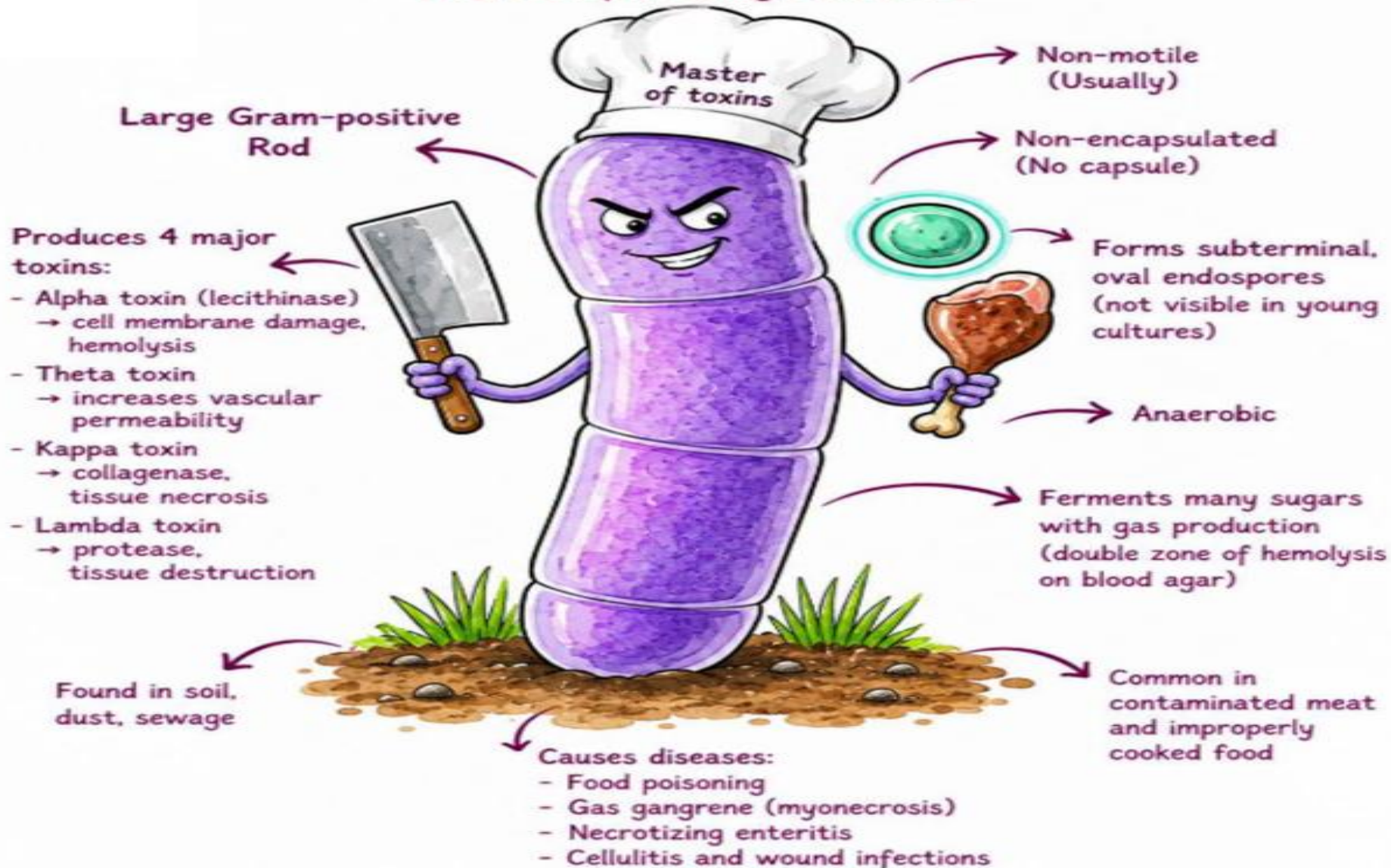


- *Clostridium perfringens* is a common food poisoning caused by *C. perfringens* bacteria.
- It causes diarrhea and stomach cramps. You usually get it from eating contaminated meat or poultry that's been left out for too long.
- *C. perfringens* food poisoning usually isn't serious and goes away on its own



Clostridium perfringens

"The food poisoning anaerobe"

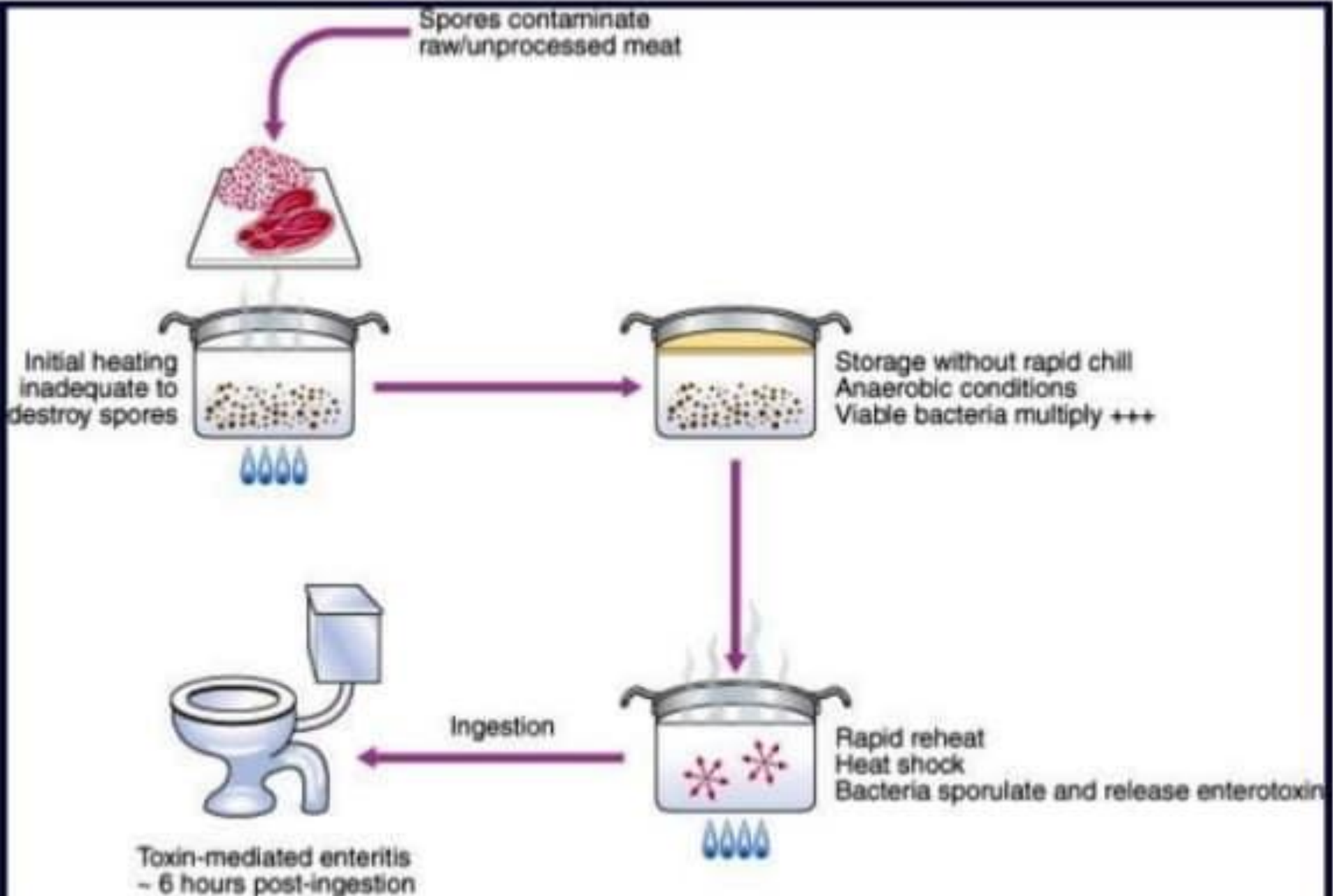


Pathogenesis :



- Spores in food may survive cooking and then germinate when they are improperly stored.
- When these vegetative cells form endospores in the intestine, they release enterotoxins.
- The bacterium is known to produce at least 12 different toxins.
- Food poisoning is mainly caused by type A strains, which produce alpha and theta toxins. The toxins result in excessive fluid accumulation in the intestinal lumen.

Pathogenesis contd..



Botulism



- These toxins are neurotoxins, that are highly toxic,
- Food contaminated with the botulism toxins heated for 100C for few minutes are safe for consumption.
- The toxins can resist the action of the gastric and intestinal juices.
- **Botulinus toxin is one of the most lethal poisons known.**
- The calculated lethal dose for an adult person is 10 µg.

Signs and Symptoms:

- Dysphagia , Diplopia , ptosis , Dysarthria , Blurring of the vision muscle weakness & even quadriplegia.
- Fever is generally retained.
- The condition is generally fatal .
- Death occurs 4-8 days later due to respiratory or cardiac failure.

Mechanism of action:-

-Under suitable anaerobic condition the toxins will be performed into the foods, **Not all strains produce toxins .**

- Food born botulism is not an infection but an intoxication since it results from the ingestion of foods that contain performed Clostridial toxin.

-It act on the parasympathetic nervous system. Its action on GI- tracts is very slight.

-It acts by blocking nerve function through inhibition of neurotransmitter acetyl choline release from the pre-synaptic membrane of neuromuscular junctions in the SNS.

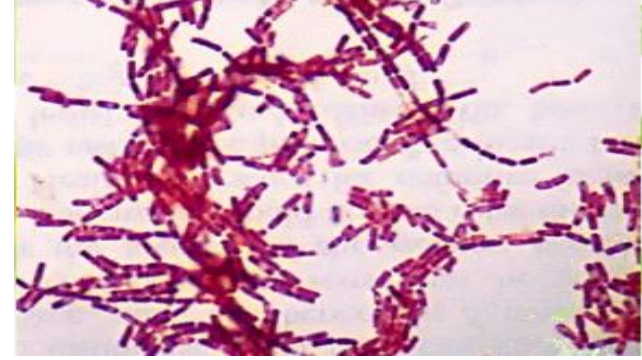


INFANT BOTULISM:



- The botulism occurring in the infants less than 6 months of age.
- It is due to the infection of the gut by the *Cl.botulinum*, with subsequent in vivo production of the toxins. Toxins are then absorbed into the bloodstream and taken throughout the body causing paralysis

Bacillus cereus



- Large motile G+Ve rod ,ubiquitous in soil & in raw,dried and prossessed foods.
- Heat resistant spores(can survive cooking , germinate & multiply rapidly when the food is held at favorable temperature.

Cereus enterotoxins:-

B-cereus produce at least two distinct enterotoxins causing two distinct form of the food poisoning.

- 1- Emetic form.
- 2- Diarrheal form.

DIARRHEAL FORM



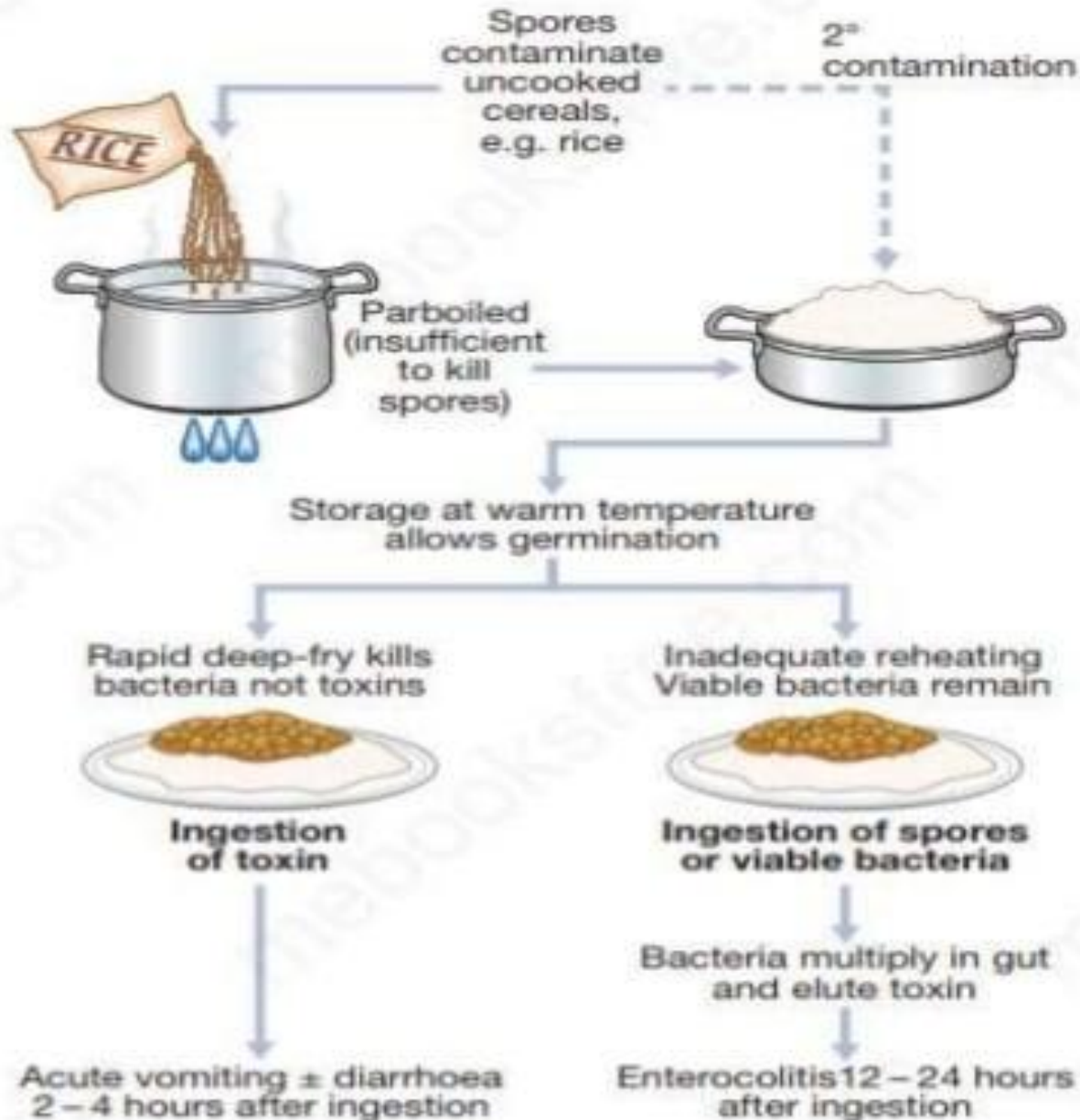
- Heat labile
- Secreted in intestine
- 8-16 hrs, incubation time
- Peristaltic movement↑
- Diarrhea, Abdominal pain
- Duration of illness is 12-24 hrs
- Meat, milk, vegetables, sauces
- Similar to C.perfringens's toxin

EMETIC FORM



- Heat stable
- Preformed
- 1-6 hrs, incubation time
- Anti-peristaltic movement↑
- Nausea, vomiting, malaise
- Duration of illness is 6-24 hrs
- Rice
- Similar to enterotoxin

Pathogenesis

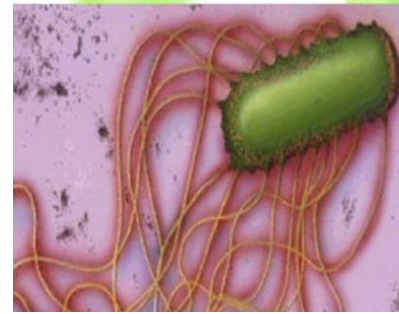


Salmonella

There are > 4000 salmonella serotypes

Most common salmonella spps

- ▶ *Salmonella* Enteritidis
- ▶ *Salmonella* Typhi



- ▶ Source of infection by asymptomatic carriers, contaminated food & water, contaminated poultry animals & products.
- ▶ Route of infection is faeco-oral route.

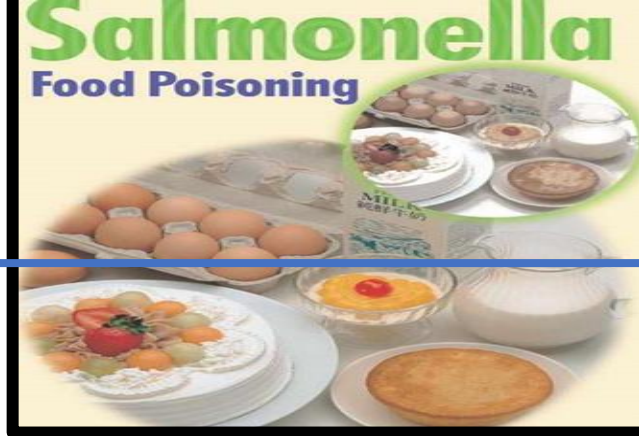
▶ Two types of salmonella infections:

- 1) Systemic infections (enteric fever): *S. Typhi*, *S. Paratyphi* A – C
- 2) Gastroenteritis (salmonellosis): *S. enteritidis*

▶ Pathogenesis :

with the invasion of intestinal epithelia

1. invasion continues and infection becomes generalized → little or no diarrhea, but pronounced fever & other general symptoms
2. infection is localized to ileocaecal region → diarrhea, nausea & vomiting, abdominal pain, temperature may be elevated



- Salmonella are **killed by pasteurization.**
- They can remain **alive in moist earth for one year and in dry earth for 16 months.**
- They are not destroyed in carcasses or offal maintained at chilling or freezing temperatures, or in the usual pickling solutions.
- S. enteritidis present inside the yolk of the egg.
- Convalescent carrier state may last for several weeks.
- The mortality rate is 1%.
- Rat & mice are another source , they are often heavily infected and contaminate the foodstuffs by their urine& faces.

Intestinal invasive infection with Salmonella Patho- physiology

Salmonella when ingested



**Penetrate & passes through EP cell lining the terminal portion of
small intestine**



Multiply in lamina propria



**Produce inflammatory mediators , recruits neutrophils
& triggers inflammation**



Release of LPS causes fever



**Inflammation causes release of prostaglandin from EP cells , cause
electrolytes to flow into lumen in response to osmotic imbalance
resulting in diarrhea (lasts 2-3 days)**

SALMONELLA

SYMPTOMS AND PREVENTION

SYMPTOMS

Fever, abdominal
cramps and
diarrhoea

Starts **12-72** hours
after infection

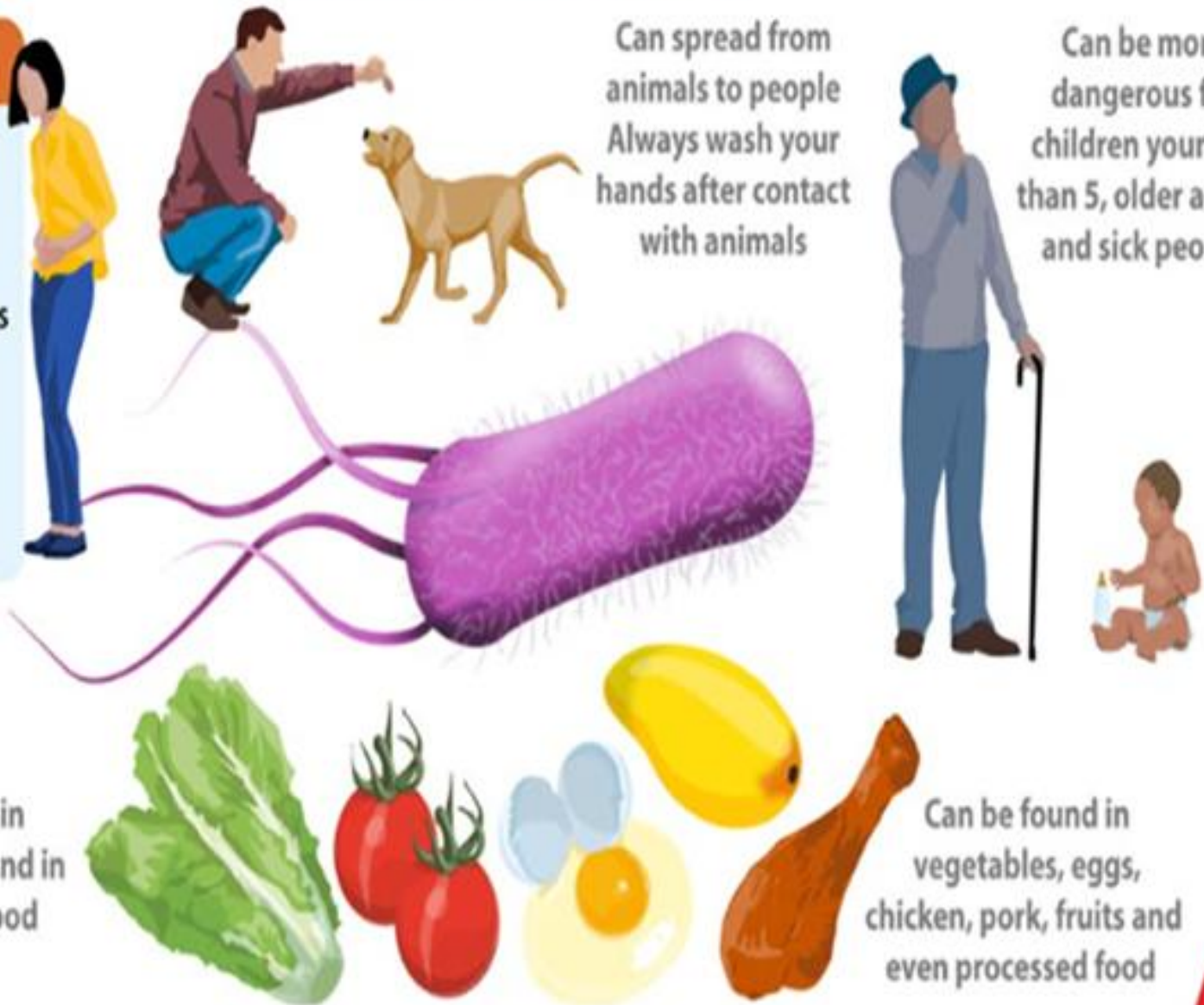
Usually lasts
4-7 days

Can spread from
animals to people
Always wash your
hands after contact
with animals

Can be more
dangerous for
children younger
than 5, older adults
and sick people

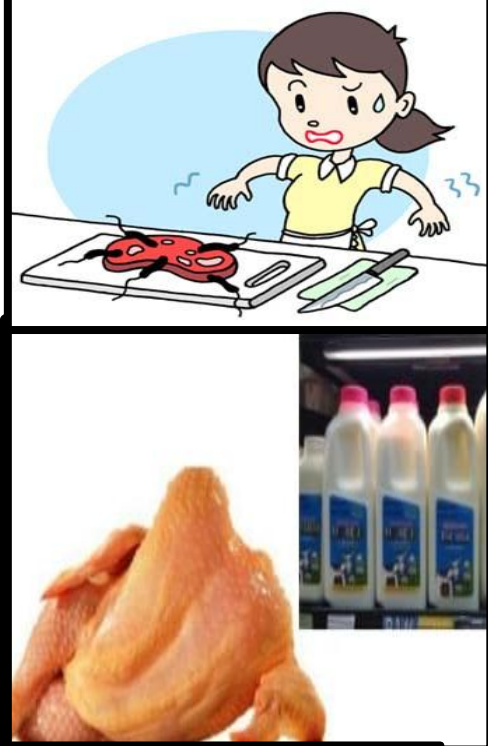
More common in
warmer weather and in
unrefrigerated food

Can be found in
vegetables, eggs,
chicken, pork, fruits and
even processed food



Campylobacter jejuni

The jejunum, ileum and colon are primarily affected resulting in acute inflammation and occasionally, abscess formation.



Clinical signs :-

- Incubation period ranges between 2-11 days with an average of 3-5 days.
- It is preceded by fever, followed by foul smelling and watery diarrhea, which runs for 3-4 days.
- The diarrhea may sometimes contain blood and mucus in feces.
- Abdominal pain is associated with backache.
- The condition is self-limiting but may last for **up to 10 days.**

Escherechia coli



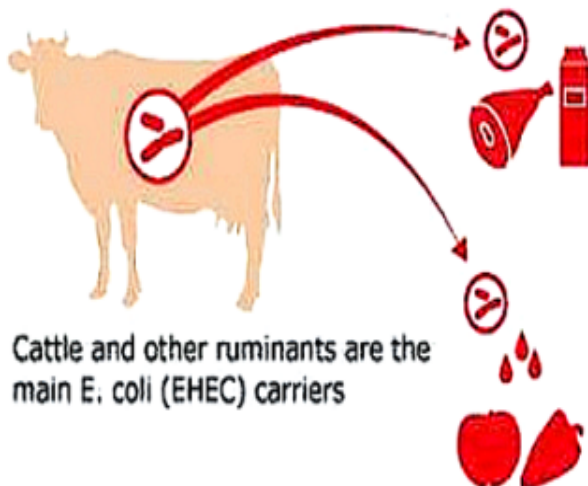
- ***E. coli* O157:H7** is responsible for several large food poisoning outbreaks.
- Traveler diarrhea mainly caused by enterotoxigenic *E.coli* .
- The presence of mucus and polymorphonuclear leucocytes in stool is typical of these strains .
- **Causes bloody diarrhea.**
- Sometimes develops into hemolytic uremic syndrome (HUS), is life threatening
- Is **heat sensitive**, but resistant to freezing.
- Illness is usually self-limiting, **lasting for 2 to 3 days.**

What are the signs and symptoms of illness caused by *E. coli* O157:H7?



Most *Escherichia coli* (*E. coli*) strains are harmless.
But some, like enterohemorrhagic *E. coli* (EHEC), are a hazard to human health and life.

INFECTION SOURCES



Uncooked meat and raw milk

The bacteria die when food is exposed to heat (70°C and higher)

Fruit and vegetables (droppings of sick animals find their way into water bodies that in turn feed the soil)

Incubation period:
three to eight days



E. coli (EHEC), once in the human stomach, begins producing toxins that cause serious illnesses

Symptoms caused by *E. coli* (EHEC)

- Stomach muscle spasms
- Diarrhea (sometimes bloody diarrhea)
- Fever
- Vomiting

Complications:

hemolytic uremic syndrome (HUS)

Death rate: 3-5 %

Cholera



- Cholera is caused by *Vibrio cholera* bacterium.
- Are ingested in drink or food. In natural infection, the dosage is usually very small.
- The organism multiply in the small intestine to produce a very potent enterotoxin, which stimulates a persistent out pouring of isotonic fluid by the gut mucosal cells.

Transmission:

- ☐ Man is the only natural host of the cholera vibrios.
- ☐ Spread of infection is from person-to-person, through contaminated water or foods.
- ☐ Shrimps and vegetables are the most frequent carriers.
- ☐ Cholera outbreaks occur as protracted epidemic waves in endemic areas.



CHOLERA

Acute, diarrheal illness caused by infection of the intestine with the bacterium *Vibrio cholerae*

Profuse watery diarrhea "rice water stools"

Vomiting

Dehydration

Circulatory collapse

Shock

Incubation period:
2 hours to 5 days

Annual Infections:
3 -5 million

Annual deaths:
100,000

Bacteria exits body through feces
Feces can be contagious for upto
14 days

Make water
safe



Prepare
safe food



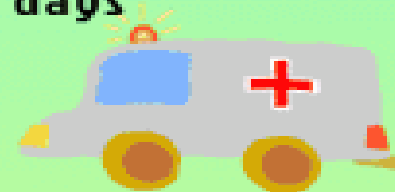
Wash your
hands



Keep
toilets
clean



Infected by
ingesting
contaminated
water or food



Clinical symptoms

- Vomiting is seen frequently, but very rapid dehydration and hypovolemic shock.
- The frequent watery stools may be accompanied with small parts of the mucosa being liberated from the intestines.
- Death may occur in 12 to 24 hrs due to rapid dehydration, because patients may produce up to 20-30 stools per day, losing many litres of water and electrolytes.
- There for exhibits extreme dehydration, urine is suppressed, the skin becomes wrinkled, the eyeballs are sunken and the voice becomes weak and husky.
- Blood pressure falls, the heart sounds become barely audible and the pulse become rapid and weak just before death.

➤ Fungal food poisoning

Mycotoxins

- Mycotoxins are fungal metabolites. Some are highly toxic to animals and potentially toxic to human beings.
- Recently concern is related to their **carcinogenic properties** and their presence in many food items.

➤ **Most are highly stable , not destroyed in normal cooking procedures(difficult to destroy by traditional food processing).**

- **Poor dry storage of grains and other foods leads to mould growth and production of mycotoxins.**

Aflatoxins



- Aflatoxicosis is caused by aflatoxins produced by the fungi, e.g. *Aspergillus flavus* & *A. parasiticus*.
- The toxin is isolated from many consumables includes rice, potato, peas etc.
- In 2003, 120 people died in Kenya after eating maize with very high aflatoxin levels.
- Four types of aflatoxins have been described i.e. aflatoxin B₁, B₂, G₁ and G₂.
- Animals consuming feeds contaminated with aflatoxin B₁ leads to secretion in the milk of aflatoxin M₁ and M₂.

High- level aflatoxin exposure produce an acute hepatic necrosis, resulting later in cirrhosis of the liver. Acute hepatic failure is made manifest by hemorrhage edema ,alteration in digestion, changes to the absorption and/ or metabolism of nutrients, and mental changes and/or coma.

Mushroom poisoning

- **Mushroom poisoning** (also known as **mycetism**) refers to harmful effects from ingestion of toxic substances present in a mushroom.
- These symptoms can vary from slight gastrointestinal discomfort to death.
- Mushroom poisoning is usually the result of ingestion of wild mushrooms after misidentification of a toxic mushroom as an edible species. The most common reason for this misidentification is close resemblance in terms of colour and general morphology of the toxic mushrooms species with edible species.
- *A. phalloides* is one of the most poisonous of all known toadstools. It has been involved in the majority of human deaths from mushroom poisoning.



7 TIPS TO AVOID FOOD POISONING

1 CHECK
Best Before date

When in doubt

**2 THROW
IT OUT**



**3 Cook to the right
TEMPERATURE**

Poultry
165°F

Ground
Meats
160°F

Whole meats
145°F



4 REFRIGERATE

Food should be refrigerated or
frozen within 2 hours on a hot day

5 CLEAN
everything in your
kitchen with hot and
soapy water

- ✓ Utensils
- ✓ Cooking surfaces
- ✓ Cutting board
- ✓ Hands



6 SEPARATE

raw meats from cooked foods when shopping
or preparing foods

7

STAY INFORMED

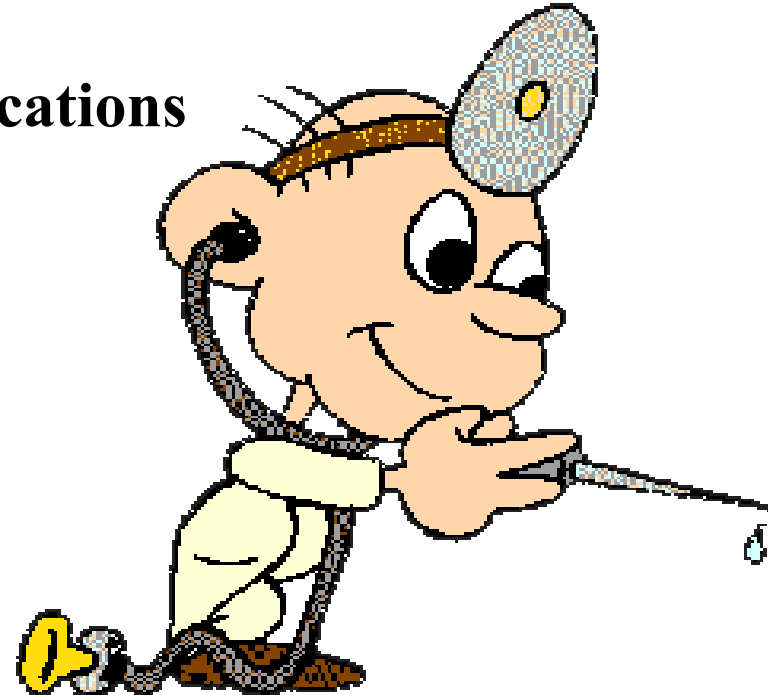
Prevention

Ultimately, prevention depends on **educating food handlers** about proper practices in cooking and storage of food and personal hygiene.



Treatment

- The main treatment for food poisoning is **putting fluids back in the body (rehydration)** through an IV and by drinking.
- Do **not eat solid food** while nauseous or vomiting but drink plenty of fluids.
- Anti-vomiting and diarrhea medications



References

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- Stefan Riedel, Thomas G. Mitchell, Jeffery A. Hobden, Judy A. Sakanari. Steve Miller, Peter Hotez, Stephen A. Morse, Rojelio Mejia, Timothy A. Mietzner. **2019. Jawetz, Melnick, & Adelberg's Medical Microbiology.** 28th edition. McGraw-Hill Education.
- Patrick R. Murray, Ken S. Rosenthal, Michael A. Pfaller. **2020. Medical Microbiology.** Elsevier

**Thank you
That's ALL**



**Would you like it with extra botulism and
a touch of salmonella ???**